A control and embedded software system for a bioinspired hummingbird *flapping-wing micro aerial vehicle.*

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SUB-SYSTEM
Control & Embedded Software
Mechanical airframe out of scope

RELEASE
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github.com/natozenbilek/hummingbird-fwmav

§ 01 — THE SYSTEM

From a Jetpack Compose touch screen to a brushed motor in four hops, with three safety nets between them.

The handset acts as the BLE central; the ESP32-C3 acts as the peripheral. A 9-byte CRC-protected command frame travels left-to-right at **50 Hz**; a 20-byte telemetry frame travels back at **10 Hz**. The firmware drives a DRV8833 H-bridge in sign-magnitude PWM at 20 kHz, 10-bit duty. Three independent fail-safes — command watchdog, BLE disconnect, CRC mismatch — gate any non-zero motor duty.



50 Hz	500 ms	150 ms	α = 0.98	9 + 20 B	3 ×
CONTROL LOOP	COMMAND WATCHDOG	HANDSET HEARTBEAT	COMP. FILTER	CMD / TLM + CRC-8	LAYERED FAIL-SAFES

Positioning vs. related work

§ II · RELATED

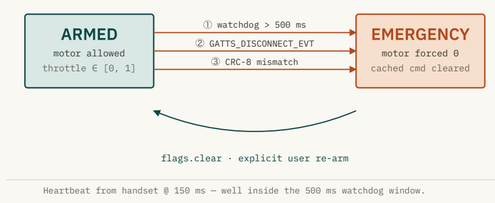
Existing hummingbird-scale FWMAVs publish flight dynamics but rarely the control-link contract or its safety behaviour. ESP32-class flight controllers do the opposite. The differentiating axes here are an explicit three-layer fail-safe and a CC0-licensed stack with an out-of-tree host-side regression target.

PLATFORM	YEAR	CMD LINK	LOSS-OF-LINK POLICY	CODE
Nano Hummingbird [1]	2012	Custom RF	Proprietary	closed
DelFly tailless [2]	2018	2.4 GHz	Disable on loss	partial
RoboBee [4]	2013	Tethered	N/A (tether)	closed
ESP32 + MAVLink [8]	2012	UDP / TCP	Heartbeat timeout	open
ESP32 BLE FC [9]	2021	BLE writes	Per-implementation	closed
This work	**2026**	**BLE GATT + CRC-8**	**Watchdog + Disconnect + CRC**	**CC0**

Three-layer fail-safe

§ IV · SAFETY

The motor receives non-zero duty only when *all three* guards agree. Any one of them, alone, forces `EMERGENCY`.



HEARTBEAT	150 ms
WATCHDOG	500 ms
CRC POLY	0x07 · x^8+x^2+x+1
RE-ARM	explicit user input

Command frame

TAB. II

9-byte little-endian frame, written without response on the command characteristic.

0	1	2	3	4	5	6	7	8

OFF	FIELD	TYPE
0	**roll** setpoint	i16 ·deg×10
2	**pitch** setpoint	i16 ·deg×10
4	**yaw** trim	i16 ·deg×10
6	**throttle**	u8 ·%
7	**flags** · e-stop / clear / rise	bitfield
8	**CRC-8** over bytes 0–7	poly 0x07

Telemetry frame

TAB. III

20-byte frame, notified to the central at 10 Hz on the bench target.

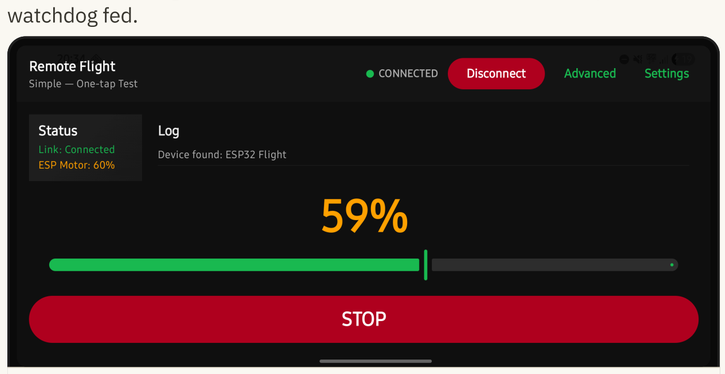
0	2	4	6	8	10	12	13	14	19

OFF	FIELD	TYPE
0	**roll** · deg×10	i16
2	**pitch** · deg×10	i16
4	**servo** roll / pitch / yaw cmds	3 × i16
10	**yaw** · deg×10	i16
12	**throttle** applied	u8 ·%
13	**flags** · e-stop	u8
14	**loop** counter	u32
18	**link** status	u8
19	**CRC-8** over bytes 0–18	poly 0x07

Handset · Simple mode

FIG. 2

Kotlin · Jetpack Compose · Material 3. Locked landscape, dark theme. Slider drives a single normalized throttle; 150 ms heartbeat keeps the watchdog fed.



SIMPLE Slider 59% · firmware echoes 60% on telemetry · 150 ms heartbeat. *Advanced* mode adds a Mode 2 dual-joystick with live telemetry cards.

Control mathematics

§ V · CONTROL

A · IMU VALIDATION

Reject any field that is not finite, |a| > 4g, or |ω| > 1000°/s — an out-of-range raw is more likely a wire fault than a maneuver. A failed validation transitions to `EMERGENCY` directly.

B · FIRST-ORDER COMPLEMENTARY FILTER

$$\phi_{acc} = atan2(-a_x, \sqrt{(a_y^2 + a_z^2)})$$

$$\theta_{acc} = atan2(a_y, a_z)$$

$$\phi_k = \alpha \cdot (\phi_{k-1} + \omega_x \cdot \Delta t) + (1-\alpha) \cdot \phi_{acc}$$

α = 0.98 trades gyro drift against accelerometer noise; yaw is integrated directly from ω_z with no heading reference at this form factor.

C · PER-AXIS PID, INTEGRAL CLAMPED AT EVERY STEP

$$u_k = k_p\, e_k + k_i \int e\, dt + k_d\, (e_k - e_{k-1}) / \Delta t$$

K_P^{ϕ} · ROLL	0.90
K_D^{ϕ} · ROLL	0.15
K_P^{θ} · PITCH	0.80
K_D^{θ} · PITCH	0.12
K_I · DEFAULT	0 · enabled by gain
\|∫e\| CLAMP	± I_{max} each step
SERVO CLAMP	± 30°

D · PITCH-TO-THROTTLE MIX

$$t_{cmd} = clamp(t_{base} - g_{mix} \cdot u_\theta,\ 0,\ 1),\ \ g_{mix} = 0.005$$

A *rise* flag temporarily sets t_{base} = 0.80 for 3 s.

Three-layer validation

§ VI · VALIDATION

Same control-math source. Three environments. Identical four-test regression in two of them; physical hardware in the third.

A — Host-side POSIX unit tests
`cmake -S . -B build && ./build/run_control_tests`. Covers clamp boundaries, IMU validation, complementary filter (zero rotation → no drift), and PID integral wind-up clamp.
● ALL 4 TESTS PASSED

B — Embedded self-test under QEMU
Same source tree, ESP-IDF v5.3, Xtensa target on QEMU. `run_control_math_self_test()` runs at firmware boot; aborts on failure so toolchain regressions are caught before reaching the C3.
● PASSES ON EVERY CI RUN

C — Physical bench demonstration
Hacettepe BBM lab, 2026-05-11. Handset discovers `ESP32 Flight`, GATT link up, brushed DC motor driven up to 60% via slider; on-board LED tracks motor state; telemetry returns within one control cycle.
● PIPELINE CONFIRMED END-TO-END

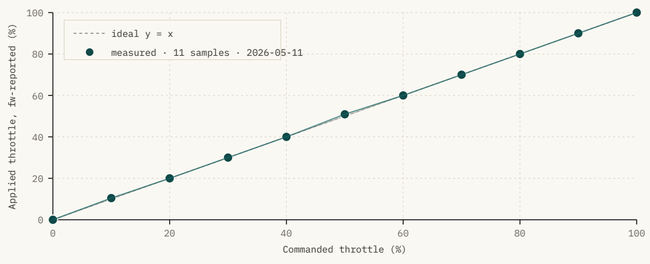


A · DC motor under finger · B · DRV8833 H-bridge · C · ESP32-C3 SuperMini · D · Android (Remote Flight) 2026-05-11

Throttle echo accuracy

FIG. 4

Slider swept 0 → 100% in 10% steps; firmware-reported `motor.throttle_pct` (telemetry offset 12) plotted against commanded throttle. Residual is the firmware's integer-percent quantisation ⌊t·100⌋.



Residual deviation: ≤ 1% absolute, attributed to `⌊throttle · 100⌋` at the BLE callback. Mean abs. error = 0.18 pp.

Remaining integration · 3 tasks

§ VII · FUTURE

Before the controller is wired to a wing actuator, three named integration points remain. All three are bring-up tasks against an already-defined interface — no algorithmic changes are required.

1 — MPU-6050 I²C driver HW
Populate `imu_raw_data_t` at 100 Hz. Complementary filter & PID already consume this via a stable interface.

2 — Battery-voltage ADC HW
A free C3 GPIO + resistor divider feeds the reserved telemetry field, completing fuel-gauge readback to the handset.

3 — Wing-stroke mixer SW
Thin mapping layer from {roll, pitch, yaw} demand to one or two motor channels, depending on airframe topology (symmetric vs. differential). DRV8833 already has the second channel.

CODE · CC0

SELECTED REFERENCES

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REPRODUCE
github.com/natozenbilek/hummingbird-fwmav
ESP-IDF v5.3 · Android SDK 34 · JDK 17+
Built & demoed at Hacettepe BBM lab, May 2026.

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DEMO VIDEO